# Discussion week 5

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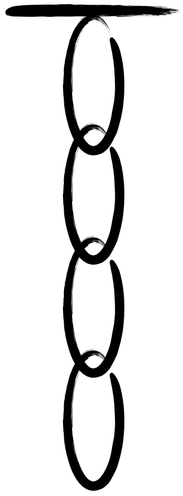
# 1.

1.A very heavy chain with four links is hanging from an elevator. Each link has mass = 1 kg. The elevator is initially at rest.



a.What are the forces between each of the links in the chain?

b.How would those forces change if the elevator was going down at a constant velocity?

c.From rest, the elevator accelerates upward at 0.5 $m/s^2$. What are the forces between each link?

a. What are the forces between each of the links in the chain?

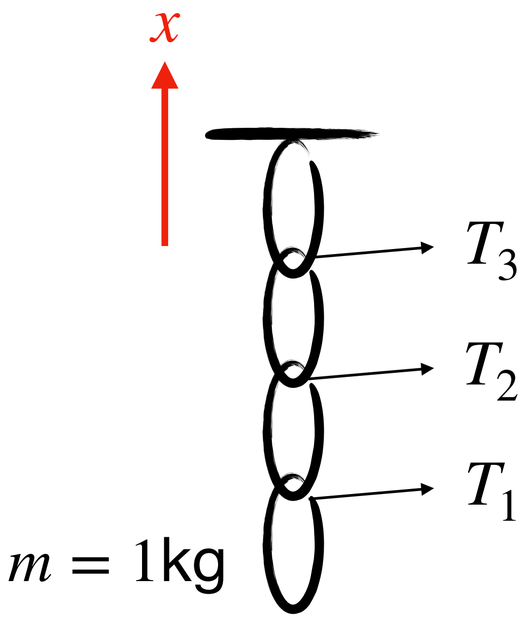
$$T_n - nmg = 0 \rightarrow T_n = nmg$$

b. How would those forces change if the elevator was going down at a constant velocity?

They would be the same.

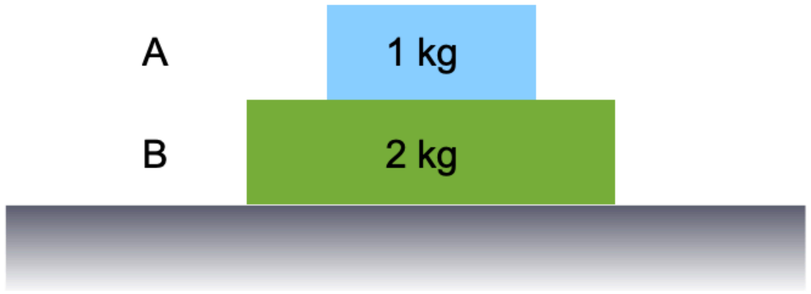
c. From rest, the elevator accelerates upward at 0.5 m/s$^2$. What are the forces between each link?

$$T_n - nmg = nma \rightarrow T_n = nm(g + a)$$



# 2.

2. Block A (1 kg) sits on block B (2 kg), which rests on a table. The coefficients of friction between the two blocks are $\mu_s$=0.5 and $\mu_k$=0.1. The coefficients of friction between block B and the table are $\mu_s$=0.7 and $\mu_k$=0.3.

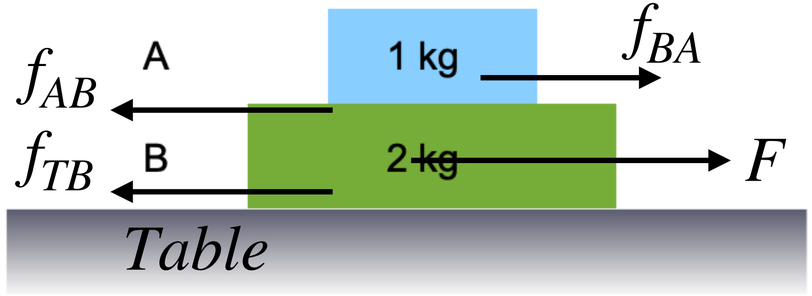



a. If I push horizontally on block B, how hard do I need to push to get it to move?
b. If I continue to push with that force on block B after it has begun moving, will block A move with the same acceleration?
c. How hard must I push on block B (horizontally) to make block A slip?

a.If I push horizontally on block B, how hard do I need to push to get it to move?

We consider A and B as a whole. Then when the system is about to slide, $F - 0.7 \times 3g = 0 \rightarrow F = 2.1g \approx 21N$.

b. If I continue to push with that force on block B after it has begun moving, will block A move with the same acceleration?




We consider A and B as a whole by assuming that A will not slide ($f_{AB} \leq f_{s,AB}^{max}$=5N). The acceleration of the whole system is $F - f_{BT} = 2.1g - 0.9g = (m_A + m_B)a \rightarrow a = 0.4g \approx 4m/s^2$. Let's check our assumption: $f_{AB} = m_A a = 4N < 5N$, so it's valid

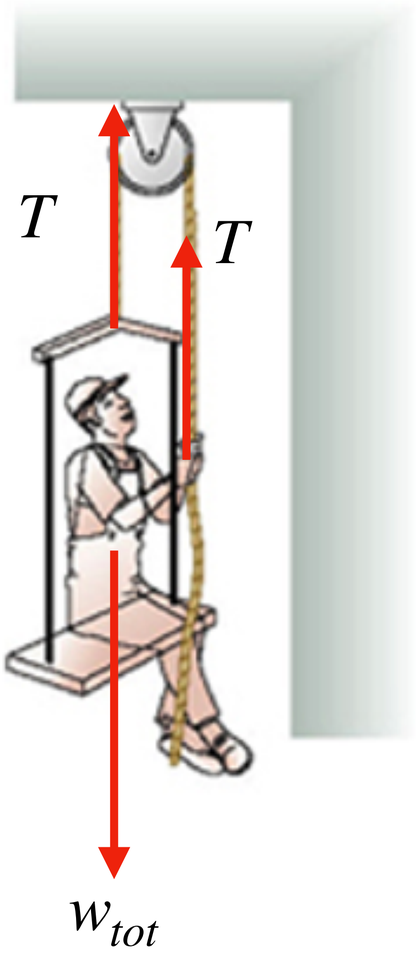
e. How hard must I push on block B (horizontally) to make block A slip?

$F\uparrow \rightarrow a\uparrow \rightarrow f_{AB}\uparrow$ A is more likely to slide

When A slides, $f_{AB} = f_{s,AB}^{max}$, so the acceleration of it is $a = \mu_s g = 0.5g$. The minimum force to make A slip happens when B is moving with this acceleration. So for the whole system, $F - fBT = F - 0.9g = (m_A + m_B) \times 0.5g \rightarrow F = 2.4g = 24N$

# 3.

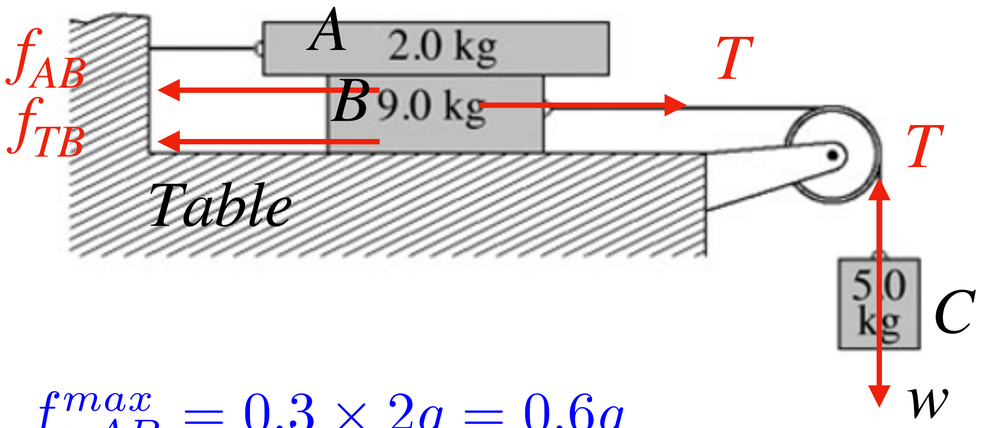
3.A house painter uses the chair and pulley arrangement of the figure to lift himself up the side of a house. The painter's mass is 70 kg and the chair's mass is 11 kg. With what force must he pull down on the rope in order to accelerate upward at 0.25 m/s$^2$




$2T - m_{tot}g = m_{tot}a$, and for massless rope the the pulling force should be equal to the tension, thus $F_{pull} = T = 1/2m(g + a)$

# 4.

4. Three blocks are arranged in the configuration shown in the figure. The coefficients of friction between all surfaces are  and . Calculate the acceleration of the 5.0-kg block.




We first calculate the maximum of static frictions. $f_{s,AB}^{max} = 0.3 \times 2g = 0.6g$ and $f_{s,TB}^{max} = 0.3 \times 11g = 3.3g$. If the system is in equilibrium, then the tension is $T = w_C = 5g > f_{s,AB} + f_{s,BT}$. Therefore, the system must move to the right. We then have Newton's 2nd Law: $m_C g - f_{k,AB} - f_{k,BT} = (m_B + m_C)a \rightarrow =$ $a \approx 1.7 m/s^2$